TATA STEEL

Attendance Reconciliation Automation System

alqama.hasnain@tatasteel.com

FMM-Infrastructure

Background & Problem Statement

The Core Challenge: CLM generates raw punch logs, but native reports fail to reconcile complex billing requirements.

Incorrect Shift Assignment

CLM systems fail to accurately map punches to scheduled shifts.

Duty vs. OT Ambiguity

Lack of clear distinction between regular duty hours and overtime.

Cross-Midnight Logic

Failure to handle complex night-shift rotations accurately.

Billing Inefficiency

Absence of skill-wise aggregates required for vendor billing.

PROJECT OBJECTIVES

Seamless Ingestion

Accepts CLM attendance data : **Presentee Report and Punch Records** via Excel (.xlsx)

Rule-Based Logic

Applies grace periods, late penalties, and OT eligibility based on employee specific profiles.

Billing Aggregates

Generates structured skill-wise and employee-wise billing reports automatically.

Work Hours
Calculation Rules

AUTOMATIC SHIFT ASSIGNMENT

Proximity Logic Overview

The system independently determines the correct shift by analyzing actual punch times, discarding CLM-assigned data.

Priority 1: Rotational

A/B/C Match within ±2 hours of start time.

Used for primary rotational shifts to ensure tight alignment with standard scheduling windows.

Priority 2: General

G/W1 evaluated via average-difference score.

Statistical evaluation for non-rotational shifts, calculating the least variance across the entire shift duration.

Includes sophisticated **Cross-Midnight Handling** for Night Shift (C).

DUTY-IN RECONCILIATION LOGIC

Scenario	Condition	Assigned Duty-In
Early OT	Arrived > 59 min early AND inOtAllowed = true	Rounded **UP** to next 30-minute mark
Grace	Arrived within 15 min late OR slightly early	**Exact** Shift Start Time
Penalty	Arrived > 15 min late	Rounded **UP** to next 30-minute mark

Example: Shift 08:30 | Arrived 08:47 → 17 min late → ***Duty In: 09:00***

DUTY-OUT RECONCILIATION LOGIC

Scenario	Condition	Assigned Duty-Out
Penalty	Left before shift end time	Rounded **DOWN** to previous 30-minute mark
OT Out	Stayed past shift end AND outOtAllowed = true	Rounded **DOWN** to previous 30-minute mark
Normal	Left at or after shift end, no OT permissions	Capped at **Exact** Shift End Time

Example: Shift ends 17:30 | Left 17:15 → Early departure → ***Duty Out: 17:00***

OVERTIME & NET CALCULATION

Total OT = In-OT + Out-OT

Floored to 0.5-hour increments

Total Working Hours = Duty Hours + OT Hours

Floored to 0.5-hour increments

Maximum Duty

Capped at 8 hours (9 with lunch). Ensures regulatory compliance.

-1

Threshold

< 30 mins work results in 0 hours of overtime payment.

Output

Automated Employee & Skill totals generated for reporting.

Architecture and Technologies Used

SYSTEM ARCHITECTURE

The tool follows a strict **Separation of Concerns** pattern, distributed across core modules:

masterFileUpload.js

Dynamic runtime data ingestion for punch logs and presentee reports.

hoursProcessing.js

Pure business logic engine for calculating attendance and OT.

renderTable.js

Multi-sheet XLSX export handler for generating billing reports.

persistent_data.js

Ground truth database storing employee profiles and master rules.

TECHNOLOGY

Lightweight & Dependency-Free

The application runs entirely in the browser, requiring no server or installation.

HTML5/CSS3

Semantic Design for accessible and responsive structure.

JS

Vanilla JS

Pure & Modular code without external frameworks

SheetJS

Robust Excel Binary I/O for handling complex data.

GitHub Pages

Reliable hosting with Zero Maintenance overhead.

EFFICIENCY

IMPACT

IMPACT ANALYSIS

BEFORE

66hrs

Manual Processing (2k employees)

Error-prone calculations, night-shift calc, and manual aggregation. Per employee ~ 2mins

→

AFTER

5sec

Automated Processing (for all)

100% accuracy, automated rounding, and one-click export.

> 99% reduction in operational processing time

DYNAMIC SCALABILITY: MASTER FILES

Runtime Database Injection

Users can upload employee details using the **Master File** Excel Sheet, or bypass, to consider all employees in CLM data.

Sheet 1: Employee Details

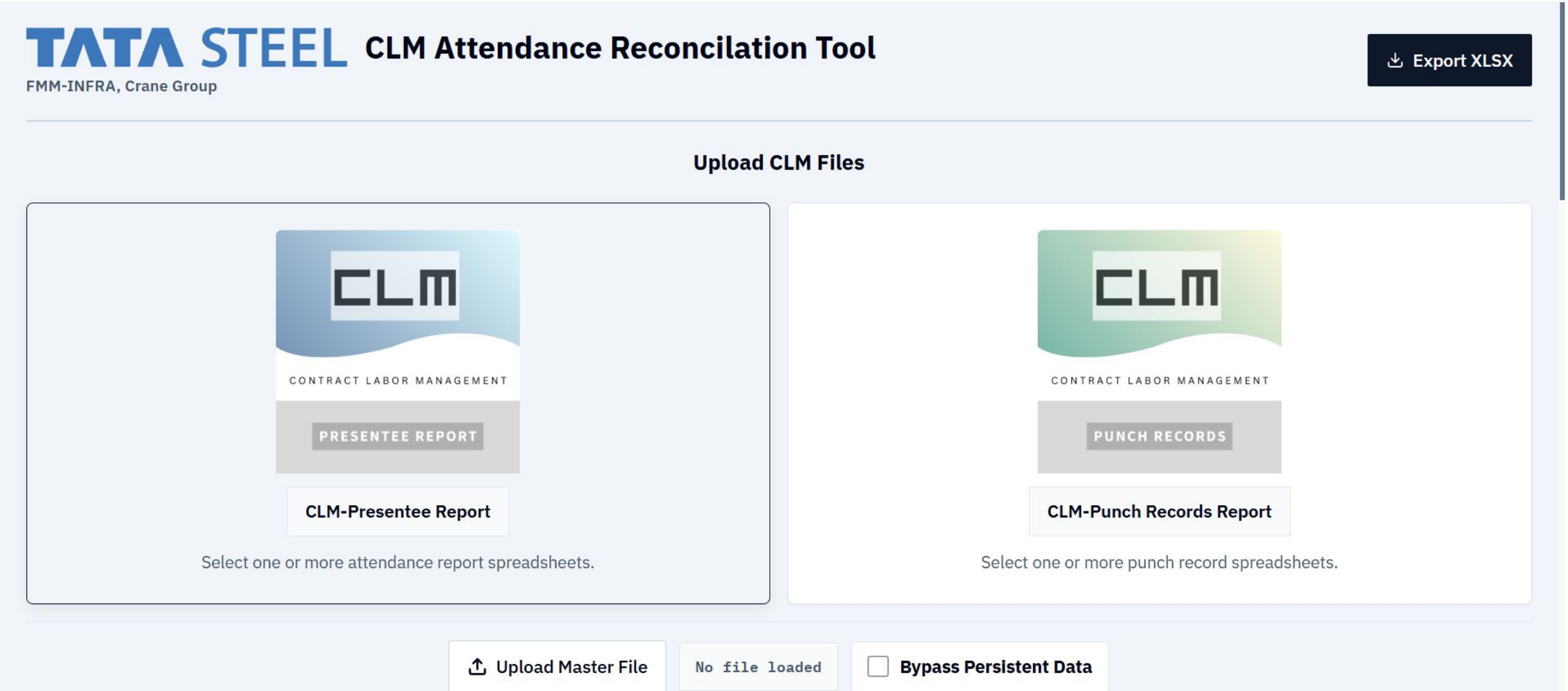
Skills, OT Permissions, and metadata

Sheet 2: Shift Definitions

Configurable In/Out times and rules

Enables seamless use across different vendor groups or departments

Demo – File Upload



Demo – Calculated Work Hours

Calculated working hours. Sortable headers are buttons and flagged shifts include text labels.

SN	DATE	SP NO	NAME	PUNCH IN	PUNCH OUT	TOTAL HRS	DUTY HRS	OT HRS	ADD LUNCH	SHIFTS ALLOWED	SHIFT	SHIFT IN	S
1	06-02-2026	RD0721287965	**SUJIT KUMAR VISHWAKARMA**	18:46	06:09	**11**	8	3	No	A, G, C	C	22:00	0
2	06-02-2026	RD0918134974	**NALIN MAHATO**	13:45	22:01	**8**	8	0	No	A, G, C	**B**	14:00	2
3	06-02-2026	RD1203000744	**RAJU KALINDI**	08:04	19:00	**9.5**	8	1.5	No	A, G, C	**W1**	08:30	1
4	06-02-2026	RS0604000979	**RAKESH KUMAR SINGH**	07:34	19:05	**10**	8	2	No	W1	**G**	08:00	1
5	06-02-2026	RW0105038642	**NAND LAL TIU**	08:00	17:37	**8**	8	0	No	W1	**G**	08:00	1
6	06-02-2026	RW0204008111	**BASUDEO**	06:13	17:21	**11**	8	3	No	A, B, C	A	06:00	1
7	06-02-2026	RW0218129961	**APURBA KUMAR MAHATO**	08:20	17:31	**8**	8	0	No	W1	W1	08:30	1
8	06-02-2026	RW0305045827	**LAXMAN BHADUK**	05:48	17:03	**11**	8	3	No	A, G, C	A	06:00	1
9	06-02-2026	RW0403000921	**MANMOHAN PRASAD**	08:37	17:31	**8**	8	0	No	W1	W1	08:30	1
10	06-02-2026	RW0512133959	**SUJIT KUMAR SINGH**	08:23	17:33	**8**	8	0	No	W1	W1	08:30	1
11	06-02-2026	RW0608113053	**RAJESH PRADHAN**	08:20	17:31	**8**	8	0	No	W1	W1	08:30	1
12	06-02-2026	RW0611151520	**V RAJA RAO**	08:30	17:31	**8**	8	0	No	W1	W1	08:30	1

Demo – Aggregation of Skill and Emp Hours

Search by name, pass, vendor, or shift...

Add Lunch | Employee Total | Skill Total | Close Aggregate View | Columns +

Calculated working hours. Sortable headers are buttons and flagged shifts include text labels.

SL.NO.	Safety Pass No	Employee Name	Total Hours	Total Shifts
1	RD0721287965	SUJIT KUMAR VISHWAKARMA	11	1.38
2	RD0918134974	NALIN MAHATO	8	1
3	RD1203000744	RAJU KALINDI	9.5	1.19

Search by name, pass, vendor, or shift...

Add Lunch | Employee Total | Skill Total | Close Aggregate View | Columns +

Calculated working hours. Sortable headers are buttons and flagged shifts include text labels.

SL.NO.	Skill Level	Total Hours	Total Shifts
1	HIGHLY SKILLED	74.5	9.31
2	SEMI SKILLED	69	8.63
3	SKILLED	48	6